Backup: open-field cross-line dimension-six terms

This note records the open-field cross-line completion of the $\langle g_s^3 G^3\rangle$ sector. These terms are not included in the compact coefficient-function appendix because their numerical effect is negligible in the accepted Borel window and the expressions are better kept in machine-readable form.

The two cross-line topologies are

$$\Pi^{ij}_{G^3,\mathrm{cross}} = \Pi^{ij}_{c2b1} + \Pi^{ij}_{c1b2}, \tag{1}$$

$$\Pi^{ij}_{c2b1} \equiv \Pi^{ij}\left[S_c^{(GG,\mathrm{open})} S_b^{(G,\mathrm{open})}\right], \qquad \Pi^{ij}_{c1b2} \equiv \Pi^{ij}\left[S_c^{(G,\mathrm{open})} S_b^{(GG,\mathrm{open})}\right]. \tag{2}$$

After the spin–1 projection and direct Borel transform, each term is evaluated as

$$\Pi^{ij}_X(M^2, s_0) = \int_{\xi_-}^{\xi_+} d\xi\, I^{ij}_X(\xi, M^2), \qquad X \in \{c2b1, c1b2\}. \tag{3}$$

The explicit I^{ij}_X are stored in the following files:

Channel	Topology	Machine-readable integrand
AA	$c2b1$	`d6_chunks/AA_c2b1_fast_borel_integrand.wl`
AA	$c1b2$	`d6_chunks/AA_c1b2_fast_borel_integrand.wl`
AB	$c2b1$	`d6_chunks/AB_c2b1_fast_borel_integrand.wl`
AB	$c1b2$	`d6_chunks/AB_c1b2_fast_borel_integrand.wl`
BB	$c2b1$	`d6_chunks/BB_c2b1_fast_borel_integrand.wl`
BB	$c1b2$	`d6_chunks/BB_c1b2_fast_borel_integrand.wl`

Explicit cross-line coefficient functions

We factor the cross-line integrands as

$$I^{ij}_X(\xi, M^2) = \langle g_s^3 G^3\rangle\, e^{-\bar{s}(\xi)/M^2}\, D^{ij}_X(\xi, M^2), \qquad X \in \{c2b1, c1b2\}. \tag{4}$$

The explicit coefficient functions are written below in terms of numerators Q^{ij}_X. This is the same information as the compact machine-readable files, but split across lines for readability.

$$
\begin{aligned}
D^{AA}_{c2b1} &= -\frac{Q^{AA}_{c2b1}}{9216\pi^2 M^8(\xi-1)^3\xi^3}, & D^{AA}_{c1b2} &= \frac{Q^{AA}_{c1b2}}{18432\pi^2 M^8(\xi-1)^3\xi^3},\\
D^{AB}_{c2b1} &= -\frac{Q^{AB}_{c2b1}}{9216\pi^2 M^8(\xi-1)^4\xi^4(m_b+m_c)}, & D^{AB}_{c1b2} &= \frac{Q^{AB}_{c1b2}}{18432\pi^2 M^8(\xi-1)^4\xi^4(m_b+m_c)},\\
D^{BB}_{c2b1} &= -\frac{Q^{BB}_{c2b1}}{9216\pi^2 M^8(\xi-1)^4\xi^4(m_b+m_c)^2}, & D^{BB}_{c1b2} &= \frac{Q^{BB}_{c1b2}}{18432\pi^2 M^8(\xi-1)^4\xi^4(m_b+m_c)^2}.
\end{aligned} \tag{5}
$$

$$
\begin{aligned}
Q^{AA}_{c2b1} = {} & 2m_b^2(\xi-1)\xi^2\left[6M^4(\xi-1)^2\xi + M^2 m_c^2(-6\xi^2+3\xi+8) + 3m_c^4(3\xi-1)\right]\\
& + 3m_b m_c\xi^2\left[-24M^4(\xi-1)^2 + 4M^2 m_c^2(\xi^2-3\xi+2) + m_c^4\xi\right]\\
& - 2\xi^3\left[-6M^6(\xi-1)^3\xi + 6M^4 m_c^2(\xi-1)^2\xi + M^2 m_c^4(-3\xi^2+3\xi+5) + 3m_c^6\xi\right]\\
& + 6m_b^4(\xi-1)^2\xi\left[M^2(\xi^2-1) + m_c^2(2-3\xi)\right]\\
& - 3m_b^3 m_c(\xi-1)\xi\left[4M^2(\xi^2-4\xi+3) + m_c^2(2\xi-1)\right] + 6m_b^6(\xi-1)^4 + 3m_b^5 m_c(\xi-1)^3,
\end{aligned} \tag{6}
$$

$$
\begin{aligned}
Q^{AA}_{c1b2} = {} & 6M^6\xi^2(\xi^2-2\xi+6)(\xi-1)^3 + 6M^4\xi(\xi^2-2\xi+6)(\xi-1)^2\left[m_b^2(\xi-1) - m_c^2\xi\right]\\
& + M^2(\xi-1)\Big[m_b^4(\xi-1)^2(3\xi^2-9\xi+1) - 24m_b^3 m_c(\xi-1)^2\xi\\
& \qquad + m_b^2 m_c^2\xi(-6\xi^3+27\xi^2-22\xi+1) + 24m_b m_c^3\xi^3 + 3m_c^4(\xi-4)\xi^3\Big]\\
& + 3\Big[m_b^6(\xi-1)^4 + 2m_b^5 m_c(\xi-1)^3 - m_b^4 m_c^2(\xi-1)^2\xi(3\xi-2)\\
& \qquad - 2m_b^3 m_c^3\xi(2\xi^2-3\xi+1) + m_b^2 m_c^4\xi^2(3\xi^2-4\xi+1) + 2m_b m_c^5\xi^3 - m_c^6\xi^4\Big],
\end{aligned} \tag{7}
$$

$$
\begin{aligned}
Q^{AB}_{c2b1} = {} & m_b^3(\xi-1)^2\xi^2\left[36M^4(\xi-1)^2 + M^2 m_c^2(-72\xi^2+123\xi-56) + 9m_c^4(1-3\xi)\right]\\
& + m_b^2 m_c(\xi-1)\xi^3\left[M^4(39\xi^2-89\xi+50) + M^2 m_c^2(-48\xi^2+92\xi-47) + 6m_c^4(1-3\xi)\right]\\
& - m_b m_c^2(\xi-1)\xi^3\left[3M^4(3\xi^2-11\xi+8) + M^2 m_c^2(-36\xi^2+39\xi-8) - 9m_c^4\xi\right]\\
& + m_c\xi^4\left[22M^6(\xi-1)^2 + M^4 m_c^2(-21\xi^2+62\xi-41) + M^2 m_c^4(24\xi^2-31\xi+10) + 6m_c^6\xi\right]\\
& + 3m_b^5(\xi-1)^3\xi\left[4M^2(3\xi^2-7\xi+4) + 3m_c^2(3\xi-2)\right]\\
& + m_b^4 m_c(\xi-1)^2\xi^2\left[M^2(24\xi^2-61\xi+37) + 6m_c^2(3\xi-2)\right] - 9m_b^7(\xi-1)^5 - 6m_b^6 m_c(\xi-1)^4\xi,
\end{aligned} \tag{8}
$$

$$
\begin{aligned}
Q^{AB}_{c1b2} = & -m_b^3(\xi-1)^2\xi\left[M^4(\xi-1)^2(15\xi+52)+M^2m_c^2\xi(24\xi^2-23\xi-1)+3m_c^4\xi(3\xi-1)\right] \\
& -6m_b^2m_c(\xi-1)\xi^2\left[3M^4(\xi-1)^2(7\xi-3)+M^2m_c^2(8\xi^3+3\xi^2-18\xi+7)+m_c^4\xi(3\xi-1)\right] \\
& +m_b(\xi-1)\xi^2\left[104M^6(\xi-1)^3+4M^4m_c^2(\xi-1)^2(6\xi+13)+4M^2m_c^4\xi(3\xi^2-\xi-2)+3m_c^6\xi^2\right] \\
& +6m_c\xi^3\left[24M^6(\xi-1)^3+12M^4m_c^2(\xi-1)^2(2\xi+1)+4M^2m_c^4\xi(\xi^2+\xi-2)+m_c^6\xi^2\right] \\
& +m_b^5(\xi-1)^3\xi\left[M^2(12\xi^2-19\xi+7)+3m_c^2(3\xi-2)\right] \\
& +6m_b^4m_c\xi\left[M^2(4\xi+7)(\xi-1)^4+m_c^2\xi(3\xi-2)(\xi-1)^2\right]-3m_b^7(\xi-1)^5-6m_b^6m_c(\xi-1)^4\xi,
\end{aligned}
\tag{9}
$$

$$
\begin{aligned}
Q^{BB}_{c2b1} = & \, 2m_b^4(\xi-1)^2\xi^2\left[6M^4(\xi-6)(\xi-1)^2+M^2m_c^2(54\xi^2-102\xi+53)+9m_c^4(2\xi-1)\right] \\
& +m_b^3m_c(\xi-1)\xi^2\left[2M^4(27\xi^3-85\xi^2+100\xi-42)+2M^2m_c^2(54\xi^2-83\xi+32)+9m_c^4(3\xi-1)\right] \\
& +2m_b^2(\xi-1)\xi^3\left[12M^6(\xi-1)^3\xi-6M^4m_c^2(2\xi^3-10\xi^2+19\xi-11)-2M^2m_c^4(27\xi^2-39\xi+17)+3m_c^6(1-4\xi)\right] \\
& -m_bm_c\xi^3\left[-4M^6(\xi-12)(\xi-1)^2+2M^4m_c^2(27\xi^3-58\xi^2+55\xi-24)+2M^2m_c^4(27\xi^2-28\xi+4)+9m_c^6\xi\right] \\
& +2\xi^4\left[12M^8(\xi-1)^4\xi-6M^6m_c^2(\xi-1)^2(2\xi^2-2\xi-5)+6M^4m_c^4(\xi^3-2\xi^2+6\xi-5)\right. \\
& \qquad \left.+M^2m_c^6(18\xi^2-18\xi+5)+3m_c^8\xi\right] \\
& -6m_b^6(\xi-1)^3\xi\left[2M^2(3\xi^2-7\xi+4)+m_c^2(4\xi-3)\right] \\
& -m_b^5m_c(\xi-1)^2\xi\left[2M^2(27\xi^2-55\xi+28)+9m_c^2(3\xi-2)\right]+6m_b^8(\xi-1)^5+9m_b^7m_c(\xi-1)^4,
\end{aligned}
\tag{10}
$$

$$
\begin{aligned}
Q^{BB}_{c1b2} = & \, 12M^8\xi^3(\xi^2-2\xi+6)(\xi-1)^4 \\
& +2M^6\xi^2(\xi-1)^3\left[3m_b^2(2\xi^3-6\xi^2-19\xi+23)-8m_bm_c(2\xi+13)-6m_c^2\xi(\xi^2-2\xi-12)\right] \\
& +2M^4\xi(\xi-1)^2\Big[3m_b^4(\xi-1)^2(\xi^2+4\xi+5)+4m_b^3m_c(8\xi^2+5\xi-13) \\
& \qquad +3m_b^2m_c^2\xi(-2\xi^3-12\xi^2+3\xi+11)-4m_bm_c^3\xi(8\xi+13)+3m_c^4\xi^2(\xi^2+10\xi+6)\Big] \\
& -M^2(\xi-1)\left[m_b^2(\xi-1)-m_c^2\xi\right]^2\left[m_b^2(18\xi^3-30\xi^2+11\xi+1)+16m_bm_c\xi-6m_c^2\xi^2(3\xi+2)\right] \\
& +3\left[m_b^2(\xi-1)-m_c^2\xi\right]^3\left[m_b^2(\xi-1)^2-m_c^2\xi^2\right].
\end{aligned}
\tag{11}
$$

At the central point $M^2 = 8.0$ GeV^2, $s_0 = 54.0$ GeV^2, the cached cross-line moments are

Channel	Π_{c2b1}	Π_{c1b2}	Sum
AA	-1.67570×10^{-6}	3.37871×10^{-7}	-1.33783×10^{-6}
AB	6.90753×10^{-7}	1.19214×10^{-7}	8.09966×10^{-7}
BB	-1.16841×10^{-6}	-7.55718×10^{-7}	-1.92413×10^{-6}

Adding these terms changes the central mixing angle by

$$\theta_{\text{with cross}} - \theta_{\text{single-line}} = -2.84 \times 10^{-4} \text{ deg}. \tag{12}$$

The corresponding relative shifts of the three Borel moments are

$$\left(\frac{\Pi^{AA}_{\text{cross}}}{\Pi^{AA}_{\text{base}}}, \frac{\Pi^{AB}_{\text{cross}}}{\Pi^{AB}_{\text{base}}}, \frac{\Pi^{BB}_{\text{cross}}}{\Pi^{BB}_{\text{base}}} \right) = (-2.84,\ -2.37,\ -4.47) \times 10^{-5}. \tag{13}$$